Effects of pistachios on body weight in Chinese subjects with metabolic syndrome

Background Studies have shown that pistachios can improve blood lipid profiles in subjects with moderate hypercholesterolemia which could reduce the risk of cardiovascular disease. However, there is also a widely perceived view that eating nuts can lead to body weight gain due to their high fat content. Purpose To investigate the impact of different dosages of pistachios on body weight, blood pressure, blood lipids, blood glucose and insulin in subjects with metabolic syndrome. Methods Ninety subjects with metabolic syndrome (consistent with 2005 International Diabetes Federation metabolic syndrome standard without diabetes) were enrolled in three endocrinology outpatient clinics in Beijing. All subjects received dietary counseling according to the guidelines of the American Heart Association Step I diet. After a 4 week run-in, subjects were randomized to consume either the recommended daily serving of 42 g pistachios (RSG), a higher daily serving of 70 g pistachio (HSG) or no pistachios (DCG) for 12 weeks. Results Subjects in all three groups were matched at baseline for BMI: DCG 28.03 ± 4.3; RSG 28.12 ± 3.22; and HSG 28.01 ± 4.51 kg/m2. There were no significant changes in body weight or BMI in any groups during the study nor any change from baseline at any time point in any group. During the entire study, there were no significant differences in waist-to-hip ratio among the groups or any change from baseline in any group (DCG -0.00 ± 0.03, RSG -0.01 ± 0.02 and HSG 0.01 ± 0.04). There were no significant differences detected among groups in triglycerides, fasting glucose and 2 hour postprandial glucose following a 75 gram glucose challenge. Exploratory analyses demonstrated that glucose values 2 h after a 75 gm glucose challenge were significantly lower at week 12 compared with baseline values in the HSG group (-1.13 ± 2.58 mmol/L, p = 0.02), and a similar trend was noted in the RSG group (-0.77 ± 2.07 mmol/L, p = 0.06), while no significant change was seen in the DCG group (-0.15 ± 2.27 mmol/L, p = 0.530). At the end of study, serum triglyceride levels were significantly lower compared with baseline in the RSG group (-0.38 ± 0.79 mmol/L, p = 0.018), but no significant changes were observed in the HSG or DCG groups. Conclusion Despite concerns that pistachio nut consumption may promote weight gain, the daily ingestion of either 42 g or 70 g of pistachios for 12

weeks did not lead to weight gain or an increase in waist-to-hip ratio in Chinese subjects with metabolic syndrome. In addition, pistachio consumption may improve the risk factor associated with the metabolic syndrome.